

LATANOST EYE DROPS 0.005%

DESCRIPTION:

LATANOST EYE DROPS 0.005%: A clear, colourless solution.

COMPOSITION:

Each 1 ml contains 50 mcg of latanoprost. Contains preservative: benzalkonium chloride 0.2 mg/ml.

PHARMACODYNAMICS:

Pharmacotherapeutic group: Ophthalmologicals; Antiglaucoma preparations and miotics, prostaglandin analogues. ATC code: S 01 E E 01

The active substance latanoprost, a prostaglandin F_{2α} analogue, is a selective prostanoid FP receptor agonist which reduces the IOP by increasing the outflow of aqueous humour. Reduction of the IOP in man starts about three to four hours after administration and maximum effect is reached after eight to twelve hours. Pressure reduction is maintained for at least 24 hours.

Studies indicated that the main mechanism of action is increased uveoscleral outflow, although some increase in outflow facility (decrease in outflow resistance) has been reported in man.

PHARMACOKINETICS:

Absorption

Latanoprost (mw 432.58) is an isopropyl ester prodrug which per se is inactive, but after hydrolysis to the acid of latanoprost becomes biologically active.

The prodrug is well absorbed through the cornea and all drug that enters the aqueous humour is hydrolysed during the passage through the cornea.

Distribution

The peak concentration in the aqueous humour is reached about two hours after topical administration.

Biotransformation and elimination

There is practically no metabolism of the acid of latanoprost in the eye. The main metabolism occurs in the liver. The half-life in plasma is 17 minutes in man. The main metabolites, the 1,2-dinor and 1,2,3,4-tetranor metabolites, exert no or only weak biological activity in animal studies and are excreted primarily in the urine.

INDICATIONS:

- Reduction of elevated intraocular pressure (IOP) in patients with open-angle glaucoma, chronic angle closure glaucoma and ocular hypertension
- Reduction of elevated intraocular pressure in paediatric patients with elevated intraocular pressure and paediatric glaucoma.

RECOMMENDED DOSAGE:

Adult (including the elderly): One drop in the affected eye(s) once daily. Optimal effect is obtained if Latanoprost is administered in the evening. The dosage of Latanoprost should not exceed once daily as it has been shown that more frequent administration decreases the intraocular pressure lowering effect.

If one dose is missed, treatment should continue with the next dose as normal. Latanost may be used concomitantly with other classes of topical ophthalmic drug products to lower IOP. If more than one topical ophthalmic drug product is being used, the drugs should be administered at least five minutes apart. Contact lenses should be removed before instillation of the eye drops and may be reinserted after 15 minutes (See WARNINGS AND PRECAUTIONS).

Paediatric: Latanost may be used in paediatric patients at the same posology as in adults. No data are available for preterm infants (less than 36 weeks gestational age). Data in age group <1 year (4 patients) are limited.

ROUTE OF ADMINISTRATION: Ophthalmic

CONTRAINDICATIONS:

Hypersensitivity to Latanoprost or to any of the excipients.

WARNINGS AND PRECAUTIONS:

Preservative:

Latanost contains benzalkonium chloride, which is commonly used as a preservative in ophthalmic products.

Benzalkonium chloride has been reported to cause eye irritation, symptoms of dry eyes and may affect the tear film and corneal surface. Should be used with caution in dry eye patients and in patients where the cornea may be compromised. Patients should be monitored in case of prolonged use.

Contact lenses may absorb benzalkonium chloride and these should be removed before applying Latanost but may be reinserted after 15 minutes

Iris pigmentation changes

Latanost may gradually change eye colour by increasing the amount of brown pigment in the iris. Before treatment is instituted, patients should be informed of the possibility of a permanent change in eye colour. Unilateral treatment can result in permanent heterochromia.

This change in eye colour has predominantly been seen in patients with mixed coloured irides, i.e. blue-brown, grey-brown, yellow-brown and green-brown. In patients with homogeneously blue eyes, no change has been observed and in patients with homogeneously grey, green or brown eyes, the change has only rarely been seen.

The colour change is due to increased melanin content in the stromal melanocytes of the iris and not to an increase in number of melanocytes. Typically, the brown pigmentation around pupil spreads concentrically towards the periphery in affected eyes, but the entire iris or parts of it may become more brownish. No further increase in brown iris pigment has been observed after discontinuation of treatment.

Neither naevi nor freckles of the iris have been affected by treatment.

Glaucoma

There is limited experience of Latanost in chronic angle closure glaucoma, open angle glaucoma of pseudophakic patients and in pigmentary glaucoma. There is no experience of Latanost in inflammatory and neovascular glaucoma or inflammatory ocular conditions. Latanost has no or little effect on the pupil, but there is no experience in acute attacks of closed angle glaucoma. Therefore, it is recommended that Latanost should be used with caution in these conditions until more experience is obtained.

There are limited study data on the use of Latanost during the peri-operative period of cataract surgery. Latanost should be used with caution in these patients.

Eyelid and eyelash changes

Latanoprost may gradually change eyelashes and vellus hair in the treated eye and surrounding areas; these changes include increased length, thickness, pigmentation, number of lashes or hairs and misdirected growth of eyelashes. Eyelash changes are reversible upon discontinuation of treatment.

Herpetic Keratitis

Latanoprost should be used with caution in patients with a history of herpetic keratitis, and should be avoided in cases of active herpes simplex keratitis and in patients with a history of recurrent herpetic keratitis specifically associated with prostaglandin analogues.

Macular Oedema

Reports of macular oedema have occurred mainly in aphakic patients, in pseudophakic patients with torn posterior lens capsule or anterior chamber lenses, or in patients with known risk factors for cystoid macular oedema (such as diabetic retinopathy and retinal vein occlusion). Latanost should be used with caution in aphakic patients, in pseudophakic patients with torn posterior lens capsule or anterior chamber lenses, or in patients with known risk factors for cystoid macular oedema.

Respiratory disorders

There is limited experience from patients with asthma, but some cases of exacerbation of asthma and/or dyspnoea were reported in post marketing experience. Asthmatic patients should therefore be treated with caution until there is sufficient experience.

Iritis/ Uveitis

In patients with known predisposing risk factors for iritis/uveitis, Latanost can be used with caution.

INTERACTIONS WITH OTHERS MEDICAMENTS:

Definitive drug interaction data are not available. There have been reports of paradoxical elevations in intraocular pressure following the concomitant ophthalmic administration of two prostaglandin analogues. Therefore, the use of two or more prostaglandins, prostaglandin analogues or prostaglandin derivatives is not recommended.

No interaction studies been performed in paediatric population.

USE IN PREGNANCY AND LACTATION:**Fertility:**

Latanoprost has not been found to have any effect on male or female fertility in animal.

Pregnancy:

There are no adequate and well-controlled studies in pregnant women. Latanoprost should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Lactation:

Latanoprost and its metabolites may pass into breast milk. Latanoprost should therefore be used with caution in nursing women.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

Instillation of eye drops may cause transient blurring of vision. Until this has resolved, patients should not drive or use machines.

SIDE EFFECTS:

The majority of adverse reactions relate to the ocular system. Other ocular adverse reactions are generally transient and occur on dose administration.

System Organ Class	Very Common	Common	Uncommon	Rare	Very Rare
Infections and infestations	-	-	-	Herpetic keratitis*#	-
Nervous system disorders	-	-	Headache*; dizziness*	-	-
Eye disorders	Iris hyperpigmentation; mild to moderate conjunctival hyperaemia; eye irritation (burning, grittiness, itching, stinging and foreign body sensation); eyelash and vellus hair changes of the eyelid (increased length, thickness, pigmentation and number of eyelashes)	Punctate keratitis, mostly without symptoms; blepharitis; eye pain; photophobia; conjunctivitis*	Eyelid oedema; dry eye; keratitis*; vision blurred; macular oedema including cystoid macular oedema*; uveitis*	Iritis*; corneal oedema*; corneal erosion; periorbital oedema; trichiasis*; distichiasis; iris cyst*#; localised skin reaction on the eyelids; darkening of the palpebral skin of the eyelids; pseudopemphigoid of ocular conjunctiva*#	Periorbital and lid changes resulting in deepening of the eyelid sulcus
Cardiac disorders	-	-	Angina; palpitations*	-	Angina unstable
Respiratory, thoracic and mediastinal disorders	-	-	Asthma*; dyspnoea*	Asthma exacerbation	-
Skin and subcutaneous tissue disorders	-	-	Rash	Pruritus	-
Musculoskeletal and connective tissue disorders	-	-	Myalgia*; arthralgia*	-	-
General disorders and administration site conditions	-	-	Chest pain*	-	-

*ADR identified post-marketing

#ADR frequency estimated using "The Rule of 3"

Cases of corneal calcification have been reported very rarely in association with the use of phosphate containing eye drops in some patients with significantly damaged corneas.

SYMPTOMS AND TREATMENT OF OVERDOSAGE:

Apart from ocular irritation and conjunctival hyperaemia, no other ocular side effects are known if latanoprost is overdosed. If latanoprost is accidentally ingested the following information may be useful: One bottle contains 125 micrograms latanoprost. More than 90% is metabolised during the first pass through liver. Intravenous infusion of 3 micrograms/kg in healthy volunteers induced no symptoms, but a dose of 5.5-10 micrograms/kg caused nausea, abdominal pain, dizziness, fatigue, hot flushes and sweating. In patients with moderate bronchial asthma, bronchoconstriction was not induced by latanoprost when applied topically on the eyes in a dose of seven times the clinical dose of latanoprost.

If overdosage with latanoprost occurs, treatment should be symptomatic.

INCOMPATIBILITIES:

Precipitation occurs when eye drops containing thiomersal are mixed with latanoprost. If such drugs are used, the eye drops should be administered with an interval of at least 5 minutes.

STORAGE CONDITIONS:

Store unopened bottle under refrigeration at 2°C to 8°C. When a bottle is opened for use, it may be stored at room temperature up to 25°C for 4 weeks. Protect from light. Keep out of reach of children. Jauhkan daripada kanak-kanak.

PACK SIZES:

Pack 1's bottle of 2.5 mL in a 5ml eye drop bottle/ box.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.